

THE HONG KONG POLYTECHNIC UNIVERSITY

RESEARCH DEGREE PROPOSAL

(To be typewritten by the Applicant)

Note: Please note that the information given in this form will only be used for processing this application.

1. Name of Proposed Chief Supervisor (if applicable):

2. Project Title:

AI-Driven Integrated Technology R&D and Clinical Translation in Biomedical Imaging and Molecular & Cellular Engineering

3. Project Objectives: (Purpose of proposed investigation)

1. Construct a closed-loop research system of "biological theory - molecular dynamics simulation - engineering technology - clinical application" to achieve in-depth integration of biomedical algorithms and engineering technologies, providing innovative technical support for biomedical research.
2. Develop an integrated biomedical analysis system incorporating automated image processing, intelligent data collection and analysis, and user-friendly interactive interfaces, lowering the threshold for clinical and research personnel to utilize AI technologies.
3. Target the research needs of neurobiology and clinical diagnosis of related diseases, develop data analysis tools based on AI and molecular dynamics simulations to improve the accuracy and efficiency of analyzing data such as fMRI and EEG, facilitating disease diagnosis and optimization of treatment plans.
4. Explore the development model of integrated software-hardware biomedical experimental equipment, create innovative tools such as intelligent biological experiment chambers, promote the rapid integration of AI technology and biotechnology, and accelerate the process of basic biomedical research.
5. Establish a virtuous cycle mechanism for algorithm optimization and clinical application feedback, iteratively upgrade algorithm models through clinical data, and promote the practical application of technical achievements in fields such as neurological disease diagnosis and drug development.

4. Scope and Background of Research:

(Please identify key issues/problems to be addressed)

Research Background

With the rapid development of biomedical engineering, interdisciplinary integration has become the core driving force for promoting breakthroughs in the field. Research in areas such as biomedical imaging and molecular & cellular engineering is increasingly dependent on advanced computer technologies, and the maturity of AI, image processing, and software development technologies has provided strong support for it. Currently, biomedical research faces numerous technical pain points: traditional data analysis relies on manual operations, which are inefficient and subjective; data interoperability between different devices is poor, making it difficult to form an integrated analysis process; the application threshold of AI technology for researchers is relatively high, limiting technology landing; algorithm models are disconnected from actual clinical needs, lacking a feedback mechanism for continuous optimization.

In the field of neurobiological research, the massive amount of data generated by technologies such as fMRI and EEG is difficult to analyze efficiently, and the exploration of key scientific issues such as neural conduction mechanisms is limited by the lack of analytical tools; in clinical disease diagnosis, early diagnosis of neurological diseases such as Alzheimer's disease and epilepsy lacks precise data analysis methods, leading to delayed diagnosis and poor treatment effects; in biomedical experiments, the lack of intelligent experimental equipment makes it difficult to achieve real-time monitoring and precise analysis of cellular activities and molecular mechanisms.

With over 10 years of experience in software algorithm development and image processing, I am proficient in programming languages such as C/C++/Python and development tools including PyTorch and OpenCV, possessing capabilities in AI model deployment, system optimization, and software-hardware integration. I have led the R&D of multiple projects such as vision-guided robotic arms and high-precision OCR recognition systems, accumulating rich engineering practical experience. Currently, I am supplementing my biomedical knowledge by self-studying courses such as Molecular Biology and Cell Biology, committed to the in-depth integration of engineering technology and biomedicine to solve the aforementioned research pain points.

Key Issues

1. Biomedical data is complex in type and diverse in dimension. How to achieve automated integration and precise analysis of data from different sources to construct a unified data analysis framework.
2. The integration mechanism between molecular dynamics simulation and AI algorithms is not yet clear. How to optimize AI data analysis models through simulating cellular and molecular activities to improve the ability to parse biological and physiological mechanisms.
3. Clinical and research scenarios have high requirements for the practicality and usability of technologies. How to balance technical advancement and user operational convenience to develop tools that truly meet actual needs.
4. Interdisciplinary cooperation faces problems such as differences in technical language and inconsistent research objectives. How to establish an efficient cooperation mechanism to ensure the collaborative advancement of engineering technology and biomedical research.
5. Clinical data involves ethical and privacy issues, with difficulties in data collection and low standardization. How to obtain high-quality data for algorithm training and verification under compliant premises.
6. The development of integrated software-hardware equipment faces challenges such as compatibility and stability. How to achieve seamless integration of functions such as imaging, reasoning, early warning, and automatic data recording.

5. Research Methodology:

1. Biological Theory Learning and Knowledge Reserve

Systematically study core courses including Neurobiology, Cell Biology, and Molecular Biology, focusing on mastering key knowledge such as visual/hippocampal neural circuits and coding mechanisms, dynamic processes of neuronal and synaptic structures, and the association between molecules and neural activities, laying a solid biomedical theoretical foundation for technological R&D. Track the cutting-edge trends of biomedical research by consulting authoritative literature in the field and participating in academic seminars to ensure that technological R&D is closely aligned with research needs.

2. Integration of Molecular Dynamics Simulation and Algorithms

Utilize molecular dynamics simulation tools such as GROMACS and LAMMPS to simulate the activities of biomolecules such as neurotransmitters, revealing cellular activities and neural conduction mechanisms. Use simulation results as supplementary training data for AI algorithms, optimize the modeling ability of algorithms for biological and physiological phenomena, realize the in-depth integration of molecular dynamics simulation and AI algorithms, and improve the scientificity and accuracy of data analysis.

3. Engineering Technology R&D

3.1 Construction of Automated Image Processing Workflow: Integrate functions such as camera image acquisition, data communication, AI reasoning, result display and feedback, and UI interaction. Develop an automated software workflow based on tools such as OpenCV and PyTorch to achieve seamless connection of various links and avoid the cumbersome process of separate operation of multiple devices.

3.2 Development of Intelligent Data Collection and Classification System: Utilize classification models (such as MobileNet architecture) and preset rules to automatically name, classify, and store collected biomedical images and data, providing standardized data support for subsequent data processing, model training and verification, and data augmentation.

3.3 Development of User-Friendly Interactive Interfaces: Develop visual operation interfaces based on tools such as QT and MFC, encapsulate complex algorithms and processing processes, design simple and intuitive operation logic, lower the threshold for medical and research personnel to use, and improve tool practicality.

3.4 AI System Deployment and Optimization: Deploy adaptive versions of AI systems according to the computing resources and platform requirements of different clinical and research scenarios to ensure the stable operation of algorithms in different hardware environments. Adopt methods such as CUDA, TBB multi-threading, and AVX instruction set optimization to accelerate model training and reasoning verification processes, and improve the speed of offline clinical reasoning.

3.5 R&D of Integrated Software-Hardware Equipment: Combine 3D printing technology, biological signal sensors, edge computing devices, etc., to develop integrated software-hardware demonstration equipment such as biological experiment chambers. Realize communication and interconnection with other biomedical equipment, integrate functions such as imaging, reasoning, early warning, automatic data recording and classification, and create intelligent experimental tools.

4. Clinical and Practical Application Verification

Application in Neurobiological Data Processing: Apply the developed image processing and AI analysis tools to fMRI and EEG data analysis, extract key features such as functional connection patterns in regions such as the visual nervous system and hippocampus, accurately identify abnormal neural activity signals, and deeply parse the pathogenesis of neurological diseases.

Assistance in Clinical Disease Diagnosis: Apply AI-optimized image and neural data analysis tools to preclinical and clinical research of diseases such as Alzheimer's disease and epilepsy, identify abnormal neural activity patterns, quantify pathological changes, and provide data support for disease diagnosis and treatment plan optimization.

Data Feedback and Algorithm Iteration: Collect data and usage feedback from clinical and research applications, analyze technical pain points and improvement directions, continuously optimize algorithm models and software functions, and form a virtuous cycle of "technology R&D - application verification - algorithm upgrade".

5. Interdisciplinary Cooperation and Data Management

Participate in cooperative projects with biomedical engineering teams, biological research institutes, and hospitals to obtain experimental environments, equipment, and clinical data support. Strictly follow data management specifications and ethical requirements, anonymize clinical data, and ensure the compliance and security of data collection and use. Actively participate in interdisciplinary communication meetings, cooperate in coordinating research objectives and progress, and assist in solving technical and communication problems in cooperation.

6. Project Significance and Value:

Academic Value

1. Constructs a closed-loop interdisciplinary research system of "biological theory - molecular dynamics - engineering technology - clinical application", providing an innovative paradigm for interdisciplinary integration in the field of biomedical engineering and enriching the research methods and technical paths in this field.
2. Realizes the in-depth integration of molecular dynamics simulation and AI algorithms, providing a new technical means for parsing biological and physiological mechanisms such as cellular activities and neural conduction, and helping to promote research breakthroughs in basic disciplines such as neurobiology.
3. The developed automated data analysis system and integrated software-hardware equipment provide standardized and intelligent technical tools for biomedical research, filling the gaps of existing technologies in data integration and efficiency improvement, and promoting the innovation of biomedical research methods.

Application Value

1. The developed fMRI and EEG data analysis tools can improve the accuracy and efficiency of neurological disease diagnosis, provide support for the early diagnosis and treatment of diseases such as Alzheimer's disease and epilepsy, help improve patient prognosis, and reduce medical burdens.
2. User-friendly operation interfaces and automated processes lower the application threshold of AI technology in the biomedical field, enabling more researchers and medical workers to use advanced technologies for research and clinical work, and accelerating technology landing and promotion.
3. Integrated software-hardware intelligent experimental equipment can promote the rapid integration of AI technology and biotechnology, helping researchers observe life phenomena and discover physiological mechanisms more efficiently, and accelerating the process of basic biomedical research and drug development.
4. The established algorithm iteration and feedback mechanism can continuously optimize technical achievements, promote their wide application in fields such as neurological disease diagnosis, rehabilitation treatment, and drug delivery, and provide successful cases for the technological transformation in the field of biomedical engineering.

7. Details of Any External Collaboration:

This project is a typical interdisciplinary research that requires establishing close cooperative relationships with teams in multiple fields. The specific cooperation details are as follows:

1. Cooperation with The Hong Kong Polytechnic University's Biomedical Engineering (BME) Team:

Rely on the team's academic accumulation and research resources in fields such as biomedical imaging and molecular & cellular engineering to obtain biomedical theoretical guidance and experimental design support. Cooperate in carrying out technology R&D and verification work to ensure that the research direction is closely aligned with academic frontiers and actual needs. Share laboratory equipment and experimental sites to provide basic conditions for the development and testing of software-hardware prototypes.

2. Cooperation with Biological Research Institutes: Obtain professional technical support in fields such as molecular biology and cell biology, carry out experiments related to molecular dynamics simulation, acquire basic data on cellular activities and molecular mechanisms, and provide real and reliable biomedical data support for algorithm training and verification. Cooperate in parsing biological and physiological mechanisms to ensure the scientificity and pertinence of technological R&D.

3. Cooperation with Hospitals: Establish clinical research cooperation relationships to obtain clinical data such as fMRI and EEG and patient case information (after anonymization) to provide data support for the clinical application verification of technologies. Set up pilot application scenarios in hospitals to test the performance and practicality of tools in actual clinical diagnosis, and collect doctors' feedback for technology optimization. Utilize the hospital's clinical resources to promote the clinical transformation and promotion of research results.

4. Cooperation Division of Labor: Researchers with computer and electronic technology backgrounds are mainly responsible for engineering technology R&D, algorithm design and optimization, and software-

hardware system integration; expert units in biological and medical fields (such as biological research institutes and hospitals) provide academic and resource support including biomedical theoretical guidance, experimental and clinical data support, experimental environment and equipment guarantee, as well as clinical application verification and feedback. Establish a regular communication mechanism to ensure efficient cooperation and jointly solve interdisciplinary problems in research.

In these circumstances, are there likely to be any complications associated with the publication of your thesis? Give details.

As a highly interdisciplinary research, this project may face the following complex issues during thesis publication:

1. Privacy Protection and Compliance of Clinical Data: A large amount of clinical data is needed for algorithm training and verification in the research. Although the data will be anonymized, there may still be risks of leakage of some sensitive clinical information. Ethical norms and compliance requirements for the use of clinical data vary across different regions and institutions. If the data collection, processing, and use processes do not fully comply with relevant regulations, it may face problems such as failure to pass ethical review and restricted data use rights, which will affect the presentation of research results and thesis publication. In addition, some hospitals may have concerns about the public sharing of clinical data, resulting in the inability to fully display data details in the thesis and affecting the persuasiveness of the research.

2. Differences in Evaluation Standards for Interdisciplinary Research Results: There are differences in academic evaluation standards and thesis publication requirements among fields such as biomedical engineering, computer science, and clinical medicine. For example, the engineering and technology field focuses on technological innovation and system performance, the biomedical field emphasizes experimental verification and mechanism analysis, and the clinical medicine field pays attention to clinical value and application effects. The thesis needs to meet the evaluation requirements of multiple fields simultaneously and balance the presentation of research results in different dimensions, which may face problems such as divergent review opinions and high modification difficulty, affecting the publication process.

3. Confidentiality of Technology and Data: Some core algorithms and software-hardware technologies developed in the project may have potential commercial application value, and cooperative units may have confidentiality requirements for relevant technical details and data. Excessively disclosing core technologies and sensitive data in the thesis may infringe on the commercial interests of cooperative units and trigger cooperation disputes; concealing key technologies and data will affect the academic value and reproducibility of the thesis, leading to review questions.

4. Data Sharing and Reproducibility Issues: The types of data used in interdisciplinary research are diverse, including simulation data, experimental data, and clinical data. Some data may not be publicly shared due to privacy protection, cooperation agreement restrictions, etc. This may make it difficult for other researchers to replicate and verify research results, affecting the academic credibility of the thesis and the citation rate after publication. At the same time, differences in experimental equipment and technical platforms in different fields may also challenge the reproducibility of research results.

8. Declaration of the Applicant

I wish to register for a research degree on the basis of the proposal given in this Form (GSB/1A).

I understand that, during the period of my registration with the University, I may not be a candidate for any other degree or award.

I understand that, except with the specific permission of the Research Committee, I must prepare and defend my thesis in English. (You are required to seek permission if another

language, which is considered more appropriate to the subject, is to be used in the presentation of the thesis. Please submit the justification together with this application.)

I undertake to abide by the general regulations of the University.

Signature 周相如 Date 2025.12.5
Name ZHOU Xiangru

During the application period

After completing Sections 1 to 8, the applicant should upload this form to the RPgAdmission system at <https://rpgadmission.polyu.edu.hk>.

After admission

RPg students should submit this form to the General Office of the Department after completing Sections 1 to 8.

For internal use only. Applicants should leave Sections 9 and 10 blank.

9. Endorsement by the Proposed Chief Supervisor

9a. Research Ethics/ Safety Approval

[For ethics approval, Chief Supervisor / Temporary Chief Supervisor please read the Ethical Clearance for Research or Teaching Projects or Investigations Involving Human Subjects, which are available at Section V of the Handbook for Projects and Grants at https://www.polyu.edu.hk/rio/docdrive/_sso/handbooks/HD_PG.pdf, and make sure that ethics approval is obtained if your project involves human subjects. For safety approval, please read the policy and procedures for safety approval available at the Health, Safety & Environment Office Homepage. Please attach the approval letter where appropriate.]

I confirm that approval:

	* has been Obtained	* is not required	* will be obtained before the start of the project
Human Research Ethics	☐	☐	☐
Animal Research Ethics	☐	☐	☐
Biological Safety	☐	☐	☐
Ionizing Radiation Safety	☐	☐	☐
Non-ionizing Radiation Safety	☐	☐	☐
Chemical Safety	☐	☐	☐

(* Please tick as appropriate)

9b. Research Facilities and Space

☐ I confirm, to the best of my knowledge, that adequate facilities and space are available to enable the student to conduct and complete the research programme in an efficient and safe manner.

☐ I would like to request the following additional research facilities and/or space to enable the student to conduct and complete the research programme in an efficient and safe manner:

Research Facilities	
Space (Other than the regular space provided by the Department for RPg students)	

Signature _____ Department/School _____
(^ Chief Supervisor / Temporary Chief Supervisor)

Name _____ Date _____
(^ Chief Supervisor / Temporary Chief Supervisor)

(^ Please delete as appropriate)

10. Recommendation of Head of Affiliated Department in the University

- ☐ I support this application and understand, on the basis of the Chief Supervisor's endorsement, that adequate research facilities and space are available to enable the student to conduct and complete the research programme in an efficient and safe manner.
- ☐ I support this application and agree to provide the additional research facilities and/or space, as requested by the Chief Supervisor in section 9b above, to enable the student to conduct and complete the research programme in an efficient and safe manner.

Signature _____
(Head of Department/Dean of School)

Date _____

~ The completed form should be kept by the General Office of the Department. ~